

Data Science Viva Preparation Guide

Section 1: Data Preprocessing & Foundations

1. Explain data preprocessing in detail.

Data preprocessing is a technique used to convert raw data into a clean, usable format. Raw data is often incomplete, inconsistent, and contains errors. Steps include:

- **Data Cleaning:** Handling missing values, smoothing noisy data, and resolving inconsistencies.
- **Data Integration:** Combining data from multiple sources.
- **Data Transformation:** Normalization and aggregation.
- **Data Reduction:** Reducing the volume but producing the same analytical results.

2. Explain Data Frame with Suitable example.

A DataFrame is a two-dimensional, size-mutable, and heterogeneous tabular data structure with labeled axes (rows and columns). It is the primary object in Pandas.

```
import pandas as pd
data = {'Name': ['Amit', 'Sita'], 'Age': [25, 22]}
df = pd.DataFrame(data)
```

3. What is the limitation of the label encoding method?

Label encoding assigns a unique integer to each category. The primary limitation is that it introduces an **ordinal relationship** where none exists. For example, encoding [Red: 0, Green: 1, Blue: 2] might lead a model to assume Blue > Green > Red, which is incorrect for categorical data.

4. What is the need of data normalization?

Normalization scales features to a standard range (usually 0 to 1). It is needed because features with larger scales (e.g., Salary) can dominate features with smaller scales (e.g., Age) in distance-based algorithms like KNN or Gradient Descent, leading to biased results.

5. Techniques for Handling Missing Data:

- **Deletion:** Listwise or Pairwise deletion.
- **Imputation:** Replacing with Mean, Median, or Mode.
- **Predictive Imputation:** Using regression or KNN to predict missing values.

Section 2: Outliers & Algorithms

1. Methods to detect outliers:

- **Z-Score:** Points with Z-score > 3 or < -3 .
- **IQR (Interquartile Range):** Points outside $[Q1 - 1.5*IQR, Q3 + 1.5*IQR]$.
- **Boxplots:** Visual representation of outliers.

2. Data Transformation Methods:

- **Scaling:** Min-Max scaling.
- **Standardization:** Z-score normalization.
- **Log Transformation:** To handle skewed data.

3. Algorithm to display statistics of Null values:

```
1. Load dataset (df).
2. Use df.isnull() to create a mask of boolean values.
3. Apply .sum() to count True values per column.
4. Display the result.
```

4. Algorithm to replace outlier with Mean:

```
1. Calculate Q1, Q3, and IQR.
2. Define bounds: lower = Q1 - 1.5*IQR, upper = Q3 + 1.5*IQR.
3. Calculate mean of the column (excluding outliers).
4. For each value in column:
    If value < lower OR value > upper:
        value = mean
```

Section 3: Statistics

1. What is Statistics?

Statistics is the science of collecting, organizing, analyzing, interpreting, and presenting data to make informed decisions.

2. Central Tendency:

- **Mean:** Average of data.
- **Median:** Middle value when sorted.
- **Mode:** Most frequent value.

3. Types of Variables:

- **Qualitative (Categorical):** Nominal (Gender) and Ordinal (Ratings).
- **Quantitative (Numerical):** Discrete (Count) and Continuous (Height).

4. Method for statistics:

Use `df.describe()`. It shows count, mean, std, min, 25%, 50%, 75%, and max.

Section 4: Linear Regression & Calculations

1. Linear Regression:

A supervised learning algorithm that models the relationship between a dependent variable (Y) and one or more independent variables (X) by fitting a linear equation $Y = \beta_0 + \beta_1 X + \epsilon$.

2. Calculation Example:

X	Y	X ²	XY	Y_pred (approx)
95	85	9025	8075	84.3
85	95	7225	8075	77.8
80	70	6400	5600	74.5
70	65	4900	4550	68.0
60	70	3600	4200	61.5

Note: Slope (b_1) ≈ 0.65 , Intercept (b_0) ≈ 22.5 .

SSR: Sum of Squares due to Regression (Explained variation).

SSE: Sum of Squares due to Error (Unexplained variation).

SST: SSR + SSE (Total variation).

R²: SSR / SST. Closer to 1 means best fit.

Section 5: Logistic Regression

1. Logistic Regression:

Used for binary classification. It uses the Sigmoid function $1 / (1 + e^{-z})$ to map predictions to probabilities between 0 and 1.

2. Difference:

- **Linear:** Predicts continuous values; used for regression.
- **Logistic:** Predicts probabilities/classes; used for classification.

Section 6: Text Analytics

1. Inverse Document Frequency (IDF):

Measures how important a term is. $IDF = \log(\text{Total Docs} / \text{Docs with Term})$. It penalizes common words like "the".

2. Stemming vs Lemmatization:

Stemming: Crude chopping (cries -> cri). **Lemmatization:** Uses vocabulary/morphology to return root word (cries -> cry).

Section 7: Visualization & EDA

- **Univariate:** Analysis of one variable (Histogram, Boxplot).
- **Multivariate:** Analysis of multiple variables (Scatter plot, Heatmap).
- **Violin Plot parameter:** hue is used to add another categorical variable.

Section 8: Big Data & Hadoop

1. Hadoop Daemons:

NameNode, DataNode, Secondary NameNode, ResourceManager, and NodeManager.

2. HDFS:

Hadoop Distributed File System. Designed to store very large data sets reliably and stream those data sets at high bandwidth to user applications.

3. Steps to install Scala:

1. Install JDK.
2. Download Scala binaries.
3. Set SCALA_HOME environment variable.
4. Update PATH.
5. Verify with `scala -version`.